

Phase-Continuous Frequency Line Track-Before-Detect of a Tone With Slow Frequency Variation

Sofia Suvorova , Andrew Melatos, Rob J. Evans, *Fellow, IEEE*, William Moran, *Member, IEEE*, Patrick Clearwater, and Ling Sun

Abstract—We consider optimal Bayesian detection of a slowly varying tone of unknown amplitude in situations characterized by very low signal-to-noise ratio (SNR) and a large number of measurements, as found in certain gravitational wave and passive sonar problems. We use a hidden Markov model (HMM) framework but, unlike typical HMM-based frequency line tracking methods, we develop a true track-before-detect algorithm, which does not threshold the blocked Fourier data and only considers frequency trails that have phase continuity across all HMM steps. We model the frequency and phase evolution as a phase-wrapped Ornstein–Uhlenbeck process. The resulting optimal detector is computationally efficient. The detectability improvement arising from phase continuity is characterized via comparative simulation for a mock, simplified gravitational wave search problem.

Index Terms—Harmonic analysis, hidden Markov model, coherent frequency tracking.

I. INTRODUCTION

THE detection and estimation of a possibly intermittent signal with unknown, wandering frequency with the aid of a hidden Markov model (HMM) is a classic signal processing problem [1], [2]. The HMM strategy performs effectively in a variety of settings, where the signal-to-noise ratio (SNR) is low but the sample size is large, including radar and sonar tracking

[3], telecommunications [4], and gravitational wave astronomy [5], [6]. It has been generalized to handle multiple signals simultaneously [7] and track amplitude and wavenumber as well as frequency [8], [9].

In many applications, frequency line tracking is semi-coherent, in the sense that the measurement corresponding to each data block involves a coherent matched filter, but the HMM does not track the phase of the signal from one data block to the next. This delivers computational savings but sacrifices sensitivity at a given false alarm rate, because it fails to exploit phase continuity to veto spurious HMM paths. An exception is the work of [8], in which the amplitude and phase are tracked jointly by an HMM, whose hidden variables are continuous instead of discrete, and which assumes simple, linear phase evolution during each data block to calculate the transition probabilities. Phase-continuous coherent tracking has the potential to contribute decisively to detection in those applications, where one operates near the detection limit, e.g. the search for gravitational radiation from cosmic sources [10].

In this paper we extend the standard HMM frequency line tracking framework to handle simultaneous frequency-phase track-before-detect. Standard HMM based frequency line trackers typically collect blocks of time samples of a measured signal (assumed to be a slowly varying tone in additive noise) and convert them into the frequency domain to form a set of discrete time Fourier transforms (DTFTs). A discrete time finite state Markov model is then constructed by identifying the frequency bins of the DTFT with the states of a Markov chain and creating state transition probabilities to describe the movement of the tone from a frequency bin in one DTFT to a frequency bin in the next DTFT. Finally, a hidden Markov model (HMM) is constructed via noisy observation of the Markov chain states, defined by a probability function (likelihood) that models the probability of a tone being present in any frequency bin (detection) based on the observed signal in the bin, a threshold, and the known SNR. The “best” Bayesian estimate of the trail of bins followed by the tone given the noisy measurements is then computed using a version of the Viterbi algorithm [11], based on dynamic programming.

Track-before-detect is a straightforward extension which does not involve threshold in the “detection” step but simply calculates the probability of a tone being present in every bin at each step and then uses the forward-backward algorithm [11] to

Manuscript received December 21, 2017; revised May 8, 2018 and September 19, 2018; accepted October 7, 2018. Date of publication October 22, 2018; date of current version November 6, 2018. The associate editor coordinating the review of this manuscript and approving it for publication was Dr. Wenwu Wang. This work was supported by the Australian Research Council Centre of Excellence for Gravitational Wave Discovery (OzGrav) under Grant CE170100004 and in part by the Aspen Center for Physics, which is supported by the USA National Science Foundation under Grant PHY-1607611. (*Corresponding author: Sofia Suvorova.*)

S. Suvorova and R. J. Evans are with the Department of Electrical and Electronic Engineering, University of Melbourne, Parkville, VIC 3010, Australia, and also with the Australian Research Council Centre of Excellence for Gravitational Wave Discovery (OzGrav), Hawthorn, VIC 3122, Australia (e-mail: ssuv@unimelb.edu.au; robinje@unimelb.edu.au).

A. Melatos, P. Clearwater, and L. Sun are with the School of Physics, University of Melbourne, Parkville, VIC 3010, Australia, and also with the Australian Research Council Centre of Excellence for Gravitational Wave Discovery (OzGrav), Hawthorn, VIC 3122, Australia (e-mail: amelatos@unimelb.edu.au; p.clearwater@student.unimelb.edu.au; lings2@student.unimelb.edu.au).

W. Moran is with the Department of Electrical and Electronic Engineering, University of Melbourne, Parkville, VIC 3010, Australia (e-mail: wmoran@unimelb.edu.au).

Color versions of one or more of the figures in this paper are available online at <http://ieeexplore.ieee.org>.

Digital Object Identifier 10.1109/TSP.2018.2877176

compute the most probable trail. In this way detection only occurs after the best trail has been estimated, hence the name.

The requirement for phase continuity of the “best” tone over the entire trail is a substantial advance and is treated in considerable detail in this paper. The HMM/AP frequency line tracker developed by Barrett *et al.* [8] moved part-way in this direction but used limited phase models and maintained detection thresholds.

Here we rigorously derive a true phase continuous track-before-detect algorithm for optimal detection of a randomly wandering tone in very low SNR conditions. The specific formulation is presented in the context of gravitational wave astronomy, but the methodology applies more generally. Development of analytical performance results for this new algorithm appears to be very difficult and is currently the subject of further study. Hence, in this paper, simulation-based performance diagnostics in the form of detection receiver operating characteristic (ROC) curves are presented.

The transition probabilities are generalized in Section II to incorporate phase continuity, under the assumption that the frequency and phase evolve jointly according to a phase-wrapped, Ornstein-Uhlenbeck process [12] (i.e. damped Brownian motion that is periodic in phase) within each data block. The emission probabilities are generalized in Section III to incorporate a dependence on phase as well as frequency, for the common situation of a harmonic signal with dual, independent polarizations. The performance of the algorithm is quantified via Monte Carlo simulations with additive white Gaussian noise in Section IV, where ROC curves are presented. It is demonstrated that the extended HMM performs nearly as well as a coherent matched filter which requires knowledge of the signal, while preserving its speed advantage and ability to track wandering tones efficiently; a formidable challenge for many matched filters. It retains the discrete hidden states of the standard HMM framework but compensates by refining the phase model and is therefore complementary to continuous-state methods descended from the work by [8]. It is especially well suited to the signals expected in gravitational wave astronomy [10], as discussed briefly in Section V.

II. TRANSITION PROBABILITIES

Consider an HMM, whose hidden state variable $\mathbf{q}(t) = [f(t), \varphi(t)]$ is two-dimensional, where $f(t) = (2\pi)^{-1}\dot{\varphi}(t)$ and $\varphi(t)$ denote the signal frequency and phase respectively as functions of time t . Let the system be measured at $N_T + 1$ equally spaced instants, $t_0 \leq \dots \leq t_{N_T}$, with $t_n - t_{n-1} = T_{\text{drift}}$ for all n , to produce the sequence of observations $O = \{o(t_0), \dots, o(t_{N_T})\}$. Then the probability that O arises from the hidden state sequence $Q = \{\mathbf{q}(t_0), \dots, \mathbf{q}(t_{N_T})\}$ is given by

$$P(Q|O) = \Pi_{\mathbf{q}(t_0)} \prod_{n=1}^{N_T} L_{o(t_n)|\mathbf{q}(t_n)} W_{\mathbf{q}(t_n)|\mathbf{q}(t_{n-1})}, \quad (1)$$

where $\Pi_{\mathbf{q}(t_0)}$, $L_{o(t_n)|\mathbf{q}(t_n)}$, and $W_{\mathbf{q}(t_n)|\mathbf{q}(t_{n-1})}$ denote the prior vector, emission probability matrix, and transition probability matrix respectively. In this paper, $o(t_n)$ and $\mathbf{q}(t_n)$ are quantized; cf. [8]. We maximize (1) recursively using the Viterbi algorithm and write $Q^*(O) = \arg \max P(Q|O)$ for the optimal Viterbi path [2].

A. Drift Time-Scale

An HMM of the class described by (1) takes an especially simple form, if one assumes that the frequency component $f(t)$ of $\mathbf{q}(t)$ stays in a single quantized state for the full duration of each HMM step, $t_{n-1} \leq t < t_n$. This is always possible, as long as $f(t)$ evolves continuously (i.e. it obeys a stochastic differential equation of Itô or Langevin type; see for example [13]), and T_{drift} , termed the drift time-scale, is chosen to satisfy

$$\left| \int_{t_{n-1}}^{t_{n-1} + T_{\text{drift}}} dt' \dot{f}(t') \right| < \Delta f, \quad (2)$$

where Δf is the frequency bin width.

There are multiple valid ways to choose Δf , T_{drift} , and the phase bin width $\Delta\varphi$. The choice is guided by whatever prior knowledge is available regarding the long-term behavior of $\dot{f}(t)$ and $\dot{\varphi}(t)$, as well as the data format and whether the analysis is to be conducted in the time or frequency domain. In this paper, we focus on frequency domain problems, where the observation $o(t_n)$ is defined in terms of Fourier-transformed data collected during the time interval $t_{n-1} \leq t < t_n$. Under these circumstances, one naturally takes $\Delta f = (2T_{\text{drift}})^{-1}$ (Nyquist). However, this frequency resolution implies an associated phase uncertainty equal to $2\pi T_{\text{drift}} \Delta f = \pi$, which is too large to allow reliable phase tracking. We are therefore obliged to track $f(t)$ with sub-grid accuracy. This is a well-known problem in signal processing, which can be tackled in several ways. One strategy is to combine the Fourier data stored in all the frequency bins during the interval $0 \leq t < t_n$, using the complex phase information from successive data blocks to progressively refine the frequency estimate. This is the essence of algorithms like the generalized phase interpolation estimator [14]. Although this approach appears superficially to violate the Markov assumption, it can be shown that a closely related algorithm based on updating a continuous hidden-state variable can be formulated as an HMM, e.g. the “HMM/AP” tracker described by [8]. A second strategy is to calculate transition and emission probabilities bin-by-bin using information from a single HMM step but refine the frequency-phase evolution model within each step. We adopt the latter strategy in this paper. The refined model is described in Section II-B. We take $\Delta\varphi = \pi/16$ typically as a good compromise between accuracy and processing speed.

B. Phase Continuity

We assume that $\mathbf{q}(t)$ undergoes Brownian motion during the interval $t_n \leq t < t_{n+1}$, with damping parameter γ and fluctuation amplitude σ . Then $\mathbf{q}(t)$ evolves according to the stochastic differential equations

$$\frac{df}{dt} = -\gamma f + \sigma \xi(t), \quad (3)$$

$$\frac{d\varphi}{dt} = 2\pi f, \quad (4)$$

where $\xi(t)$ is a Langevin “force” obeying the ensemble averages $\langle \xi(t) \rangle = 0$ and $\langle \xi(t)\xi(t') \rangle = \delta(t - t')$. Noise enters $f(t)$ directly through $\xi(t)$ in (3) and enters $\varphi(t)$ indirectly through the integral of $f(t)$ in (4). Equations (3) and (4) are solved analytically in Appendix. The probability density function (PDF)

$p(t, \mathbf{q})$ satisfies the backward Fokker-Planck equation

$$\frac{\partial p}{\partial t} = \gamma \frac{\partial(fp)}{\partial f} - 2\pi f \frac{\partial p}{\partial \varphi} + \frac{\sigma^2}{2} \frac{\partial^2 p}{\partial f^2} \quad (5)$$

subject to the final condition

$$p(t_{n+1}, \mathbf{q}) = \delta[\mathbf{q} - \mathbf{q}(t_{n+1})] \quad (6)$$

and periodic boundary conditions in φ . Upon Fourier transforming (5), solving for the characteristic function, and inverting the result, we show in Appendix that the PDF takes the form

$$p(t, \mathbf{q}) = (2\pi)^{-1} (\det \Sigma)^{-1/2} \times \sum_{m=-\infty}^{\infty} \exp [-(\mathbf{q} - \bar{\mathbf{q}}_m) \Sigma^{-1} (\mathbf{q} - \bar{\mathbf{q}}_m)^T] \quad (7)$$

for $t \leq t_{n+1}$, where $\mathbf{q}_m$ is a 2×1 vector, Σ is a 2×2 matrix with components, respectively

$$(\bar{\mathbf{q}}_m)_1 = f(t_{n+1}) \exp(-\gamma\tau) \quad (8)$$

$$(\bar{\mathbf{q}}_m)_2 = \varphi(t_{n+1}) - 2\pi m + \frac{f(t_{n+1})}{2} [1 - \exp(-\gamma\tau)] \quad (9)$$

$$\Sigma_{11} = \frac{\sigma^2}{2\gamma} [1 - \exp(-2\gamma\tau)] \quad (10)$$

$$\Sigma_{12} = \Sigma_{21} = \frac{\sigma^2}{2\gamma^2} [1 - \exp(-\gamma\tau)]^2 \quad (11)$$

$$\Sigma_{22} = \frac{\sigma^2}{2\gamma^3} \{1 + 2\gamma\tau - [2 - \exp(-\gamma\tau)]^2\}. \quad (12)$$

Here m is a signed integer, we have $\tau = t_{n+1} - t$ and the superscript T denotes the transpose. Equation (7) can be rewritten as a Jacobi theta function.

To compute discrete transition probabilities $W_{\mathbf{q}_{i',j'}, \mathbf{q}_{i,j}}$ from a discrete state $\mathbf{q}_{i,j} = (f_{\min} + i\Delta f, j\Delta\varphi)$ at time t_n , where $f_{\min}$ is the lower edge of the search band, into a discrete state $\mathbf{q}_{i',j'} = (f_{\min} + i'\Delta f, j'\Delta\varphi)$ at time t_{n+1} the following integrals are evaluated

$$W_{\mathbf{q}_{i',j'}, \mathbf{q}_{i,j}} = \int_{\text{cell}(i,j)} p(\tau, \mathbf{q}) d\mathbf{q}, \quad (13)$$

where $\text{cell}(i, j)$ is the rectangular cell $[(i-1)\Delta f, i\Delta f] \times [(j-1)\Delta\varphi, j\Delta\varphi]$.

III. EMISSION PROBABILITY

The emission probability depends on the particular HMM state measurement process. Moreover, enforcement of phase continuity imposes additional constraints on the likelihood calculation. In order to explain these issues in proper detail, we select the particular case of the measurement process used in gravitational wave astronomy because we simulate this example later in the paper. For other applications such as passive sonar, an analogous procedure is required.

We consider a signal $h(t)$ that is a linear combination of two polarizations,

$$h(t) = \mathbf{A}^T \mathbf{B}(t) \exp[i\varphi(t)], \quad (14)$$

with $d\varphi(t)/dt = 2\pi f(t)$, $\mathbf{A} = [A_a, A_b]$ and $\mathbf{B}(t) = [a(t), b(t)]$. In (14), $a(t)$ and $b(t)$ are known, slowly varying

(e.g. $|\dot{a}|/a \ll f$), real-valued functions, which may characterize a polarization-dependent receiver response for example, and A_a and A_b are unknown, constant, complex-valued amplitudes to be estimated below. Equation (14) is applicable in general but is written in a particular form with an eye towards applications in gravitational wave astronomy (see Section V), where $a(t)$ and $b(t)$ are antenna beam-pattern functions defined by [15], which oscillate as the Earth (and hence the detector) rotates.

The log-likelihood for measurements in the interval $t_{n-1} \leq t < t_n$ in the data stream $x(t) = h(t) + n(t)$, where $h(t)$ is given by (14) and $n(t)$ denotes additive complex Gaussian noise, is given by

$$\ln \Lambda(\mathbf{A}) = \frac{1}{2\sigma_n^2} \left(\mathbf{A} \mathbf{R}^T - \frac{1}{2} \mathbf{A} \mathbf{H} \mathbf{A}^T \right) \quad (15)$$

after discarding constants. $\ln \Lambda(\mathbf{A})$ is a function of $\mathbf{A}$, f , and φ with $\mathbf{R} = \text{Re } \mathbf{S}$ and

$$\mathbf{S} = \int_{t_{n-1}}^{t_n} \mathbf{B}(t) x(t) \exp(2\pi i f t - i\varphi) dt. \quad (16)$$

In (15) and (16), σ_n^2 is the variance of the noise (variance $\sigma_n^2/2$ in each of the real and imaginary parts of $n(t)$). The 2×2 matrix $\mathbf{H}$ in (15), whose elements are given by

$$\mathbf{H}_{11} = \frac{2}{T_{\text{drift}}} \int_{t_{n-1}}^{t_n} a(t)^2 dt, \quad (17)$$

$$\mathbf{H}_{12} = \mathbf{H}_{21} = \frac{2}{T_{\text{drift}}} \int_{t_{n-1}}^{t_n} a(t) b(t) dt, \quad (18)$$

$$\mathbf{H}_{22} = \frac{2}{T_{\text{drift}}} \int_{t_{n-1}}^{t_n} b(t)^2 dt, \quad (19)$$

describes the time-averaged polarization response. Substituting the standard, maximum likelihood, amplitude estimate for $\mathbf{A}$, $\hat{\mathbf{A}} = \mathbf{R} \mathbf{H}^{-1}$, into (15) yields the detection statistic

$$\ln L_{o(t_n) \mathbf{q}(t_n)} \propto \frac{1}{2\sigma_n^2} \mathbf{R} \mathbf{H}^{-1} \mathbf{R}^T. \quad (20)$$

Equation (20) is termed the $\mathcal{F}$ -statistic in gravitational wave applications [15] with $\mathbf{R}$ replaced by $\mathbf{S}$ in Eq. (16). It is the result of maximizing with respect to $\mathbf{A}$ only, not f and φ , assuming implicitly a uniform prior on $\mathbf{A}$. A detailed justification of the role and implementation of the prior is given in [16].

The $\mathcal{F}$ -statistic in the form (20) fails to distinguish between the phases φ and $\varphi + \pi$; there is a degeneracy with the sign of the components of $\mathbf{A}$, which causes the HMM path to bifurcate at each step.

To incorporate physical constraints on the range of values of $\mathbf{A}$, we adopt a Bayesian approach and impose a prior on $\mathbf{A}$ [16]. Specifically, we choose the prior to be uniform (i.e. Lebesgue measure) on the region specified by $A_a \geq |A_b|$. This is an improper prior, as the region has infinite measure and does not accurately reflect the physics, but it works better from a computational perspective. A more appropriate prior in gravitational wave applications would favour smaller $|\mathbf{A}|$ (e.g. $\text{PDF} \propto |\mathbf{A}|^{-1}$, though this too is improper) and assume isotropic source orientations [16]. However, detailed Monte Carlo simulations confirm that the unphysical prior and the more physically appropriate priors perform similarly, yielding false dismissal probabilities

within ≈ 5 per cent of one another for false alarm probabilities greater than 10^{-3} [16]. Using the uniform (unphysical) prior, we obtain the posterior

$$\ln L_{o(t_n)q(t_n)} = \ln \int_0^\infty dA_a \int_{-A_a}^{A_a} \Lambda(\mathbf{A}) dA_b. \quad (21)$$

In general (21) must be computed numerically; it cannot be evaluated exactly in closed form. Motivated in part by gravitational wave applications, which are computationally intensive, we evaluate $L_{o(t_n)q(t_n)}$ for a particular case, studied in the next section, where both beam-pattern functions $a(t)$ and $b(t)$ are constant. In this case, we define $w = \mathbf{A}\mathbf{B}^T > 0$ and notice that $\mathbf{A}\mathbf{H}\mathbf{A}^T = (\mathbf{A}\mathbf{B}^T)(\mathbf{A}\mathbf{B}^T)^T = w^2$, which reduces the double integral in (21) to a single integral over w .¹ Hence, (15) can be replaced by

$$\ln \Lambda(\mathbf{A}) = \frac{1}{2\sigma_n^2} \left(wX - \frac{1}{2}w^2 \right), \quad (22)$$

with

$$X = \text{Re} \int_{t_{n-1}}^{t_n} x(t) \exp(2\pi i f t - i\varphi) dt. \quad (23)$$

Upon marginalizing $\Lambda(\mathbf{A})$ with respect to a uniform prior for $w \in (0, w_{\max})$, the likelihood can be computed exactly. The result for $w_{\max} \rightarrow \infty$ is

$$\ln L_{o(t_n)q(t_n)} \propto X^2/2 + \ln \left[1 + \text{erf} \left(X/\sqrt{2} \right) \right], \quad (24)$$

where

$$\text{erf}(x) = \frac{2}{\sqrt{\pi}} \int_0^x dy \exp(-y^2) \quad (25)$$

is the error function. The second term in (24) suppresses the phase option $\varphi + \pi$.

IV. WORKED EXAMPLE: $a(t), b(t)$ CONSTANT

In this section we test the performance of the HMM for the simple situation, where the detector response functions are constant, and $\mathbf{R}$ reduces to a Fourier transform of the data stream. The tests are based on Monte Carlo simulations of synthetic signals of the form (14) with $\mathbf{B}(t) = [(1/3)^{1/2}, (2/3)^{1/2}]$ and $\mathbf{A} = h_0[2^{-1/2}, 2^{-1/2}]$, where h_0 is a signal amplitude. The signal frequency $f(t)$ ramps linearly during the interval $t_n \leq t < t_{n+1}$, with $|f(t_{n+1}) - f(t_n)| < \Delta f$, while f jumps discretely at every time-step t_n , so that $f(t_{n+1}) - f(t_n)$ is distributed uniformly in the range $[-\Delta f, \Delta f]$. Hence $f(t)$ and $\varphi(t)$ are continuous for all t . The results in this section apply for arbitrary T_{drift} . As a point of reference, however, gravitational wave applications (see Section V) typically have $f \sim 10^2$ Hz, $T_{\text{drift}} \sim 10$ d, and $T_{\text{obs}} \sim 1$ yr.

Figs. 1 and 2 summarize the output of the HMM for two synthetic signals with signal-to-noise ratio λ defined as $\lambda = h_0^2 T_{\text{obs}} / S_h(f) = 1.75$ and 1.25 respectively, where $S_h(f)$ is the one-sided power spectral density of the noise at the search frequency f , and $T_{\text{obs}} = N_T T_{\text{drift}}$ is the total observation time. The stronger signal is detectable with and without

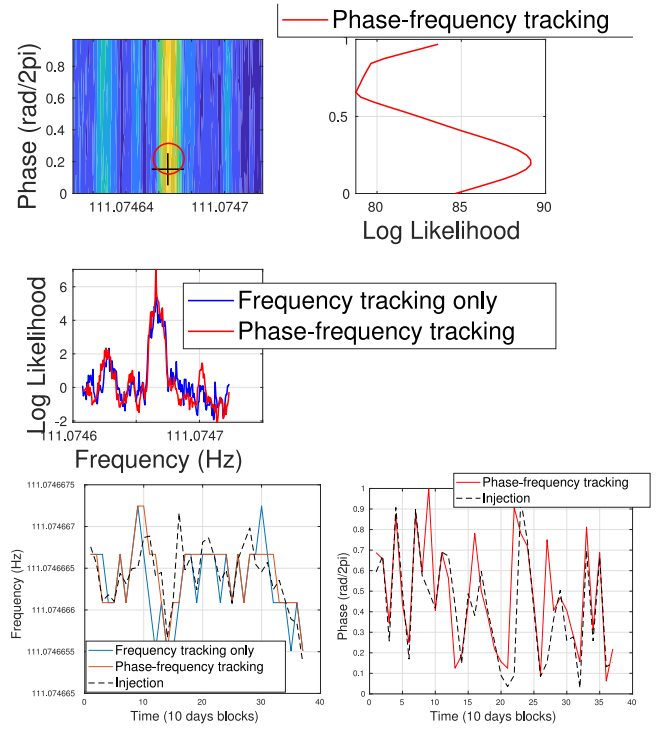


Fig. 1. Comparison of HMM performance when tracking f and φ (red curves) and tracking f only (blue curves) for a synthetic signal of the form (14) with $\lambda = h_0^2 T_{\text{obs}} / S_h(f) = 1.75$. Top left: log likelihood $\ln[Q^*(O)|O]$ of the optimal Viterbi path versus final-step frequency $f(t_{N_T})$ and final-step phase $\varphi(t_{N_T})$. Top right: contour cross-section showing log likelihood versus frequency for $\varphi(t_{N_T}) = 1.32$ rad. Middle row: contour cross-section showing log likelihood versus frequency for $\varphi(t_{N_T}) = 1.32$ rad. Bottom row: frequency component $f(t)$ (left panel) and phase component $\varphi(t)$ (right panel) of the optimal Viterbi path versus time t with (red curve) and without (blue curve) phase tracking compared to the true path (black dashed curve). The red circle and black cross correspond to the optimal Viterbi path (phase-frequency tracking) and true path respectively. Signal parameters: $\mathbf{B}(t) = [(1/3)^{1/2}, (2/3)^{1/2}]$, $\mathbf{A} = h_0[2^{-1/2}, 2^{-1/2}]$; $f(t)$ is piecewise linear and wanders by at most one bin during T_{drift} . Search parameters: $T_{\text{drift}} = 10$ d, $\gamma T_{\text{drift}} = 8.6 \times 10^{-11}$, $\sigma^2 T_{\text{drift}}^3 = 0.23$, and $N_T = 37$.

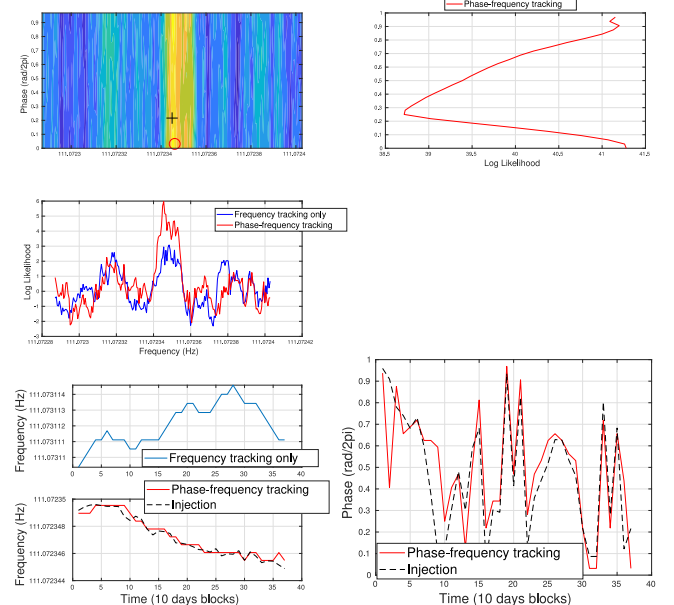


Fig. 2. As for Fig. 1 with $\lambda = 1.25$.

¹ w must be positive to resolve the phase ambiguity discussed above.

phase tracking, whereas the weaker signal is detectable with phase tracking but not without. The three topmost panels in both figures display contours of $\ln P[Q^*(O)|O]$ on the $f(t_{N_T})$ - $\varphi(t_{N_T})$ plane (top left panel), flanked below and to the right by perpendicular cross-sections through the maximum, e.g. at $[f(t_{N_T}), \varphi(t_{N_T})] = (111.07 \text{ Hz}, 1.32 \text{ rad})$ in Fig. 1. The contours are computed with phase tracking; the cross-sections are computed with (red curves) and without (blue curves) phase tracking. The bottom two panels plot the frequency and phase components of $Q^*(O)$ versus time, extracted with and without phase tracking, as well as the true evolution of the synthetic signal (black dashed curve). Note that, in Fig. 1, there is an appreciable discrepancy between the true and extracted phases (0.13 rad versus 0.21 rad respectively) at $t = t_{N_T}$, but this occurs by chance; the root-mean-square error over the whole path is still smaller than one bin. Conversely, in Fig. 2, the panel graphing log likelihood versus frequency appears to exhibit a peak in the correct place for the frequency-only HMM (blue curve), but the lower panels indicate that this too occurs by chance; the optimal Viterbi path does not match the true path, and the frequency-only HMM does not detect the signal. In all the above cases, the HMM without phase tracking tracks f using $A_{q_j q_i} = (\delta_{q_j, q_{i+1}} + \delta_{q_j, q_i} + \delta_{q_j, q_{i-1}})/3$ and $\mathbf{R}$ given by (16) with the factor $\exp(-i\varphi) = \exp[-i\varphi(t_{n-1})]$ set to unity.

At a given false alarm probability P_{fa} , the HMM with phase tracking is a more reliable detector than the HMM without phase tracking. To quantify this assertion, we construct ROC curves for both tracking modes for representative signals near the detection limit with $\lambda = 1.25, 1.5$, and 1.75 . The results are presented for comparison in Fig. 3. There are three ROC curves in each panel of the figure, corresponding to the HMM with phase tracking (red curve), the HMM without phase tracking (blue curve), and a fully coherent, semi-clairvoyant search (black dashed curve; control experiment), which involves feeding the known phase into (24) while asking the HMM to track the unknown frequency. An HMM detection is counted as successful, if $f(t_{N_T})$ for the top-ranked Viterbi path lies within $\pm N_T$ frequency bins of the true $f(t_{N_T})$ value, and the log likelihood exceeds a threshold.² A false alarm is a path, whose log likelihood exceeds the threshold, and for which $f(t_{N_T})$ lies outside the above range. Near the detection limit, at $\lambda = 1.25$, we find that the detection probability P_d achieved by the HMM with phase tracking is systematically higher than that achieved by the HMM without phase tracking by ≈ 0.10 – 0.30 for false alarm probabilities in the range $10^{-4} \leq P_{fa} \leq 10^{-2}$. A similar margin in favour of phase tracking persists for the higher λ values plotted in Fig. 3, with $P_d \geq 0.9$ for $P_{fa} \geq 2 \times 10^{-6}$ and $\lambda = 1.75$.

The performance of the phase-frequency HMM scales as expected with $\lambda \propto \text{SNR}^2$ and N_T , as illustrated by Fig. 4. Consider firstly the signal-to-noise dependence in the context of a fiducial search spanning 2×10^4 frequency bins. At a fixed false alarm probability, e.g. $P_{fa} = 5 \times 10^{-7}$ per frequency bin, and fixed $T_{\text{obs}} = 37T_{\text{drift}}$, the detection probability increases, as λ increases. Upon inspecting Fig. 4(a), we find $P_d \geq 0.9$ for $\lambda \geq 1.75$, with P_d falling away rapidly for $\lambda \lesssim 1.5$. Moreover, for fixed P_d and $\lambda \geq 1.25$, the phase-frequency HMM detects λ

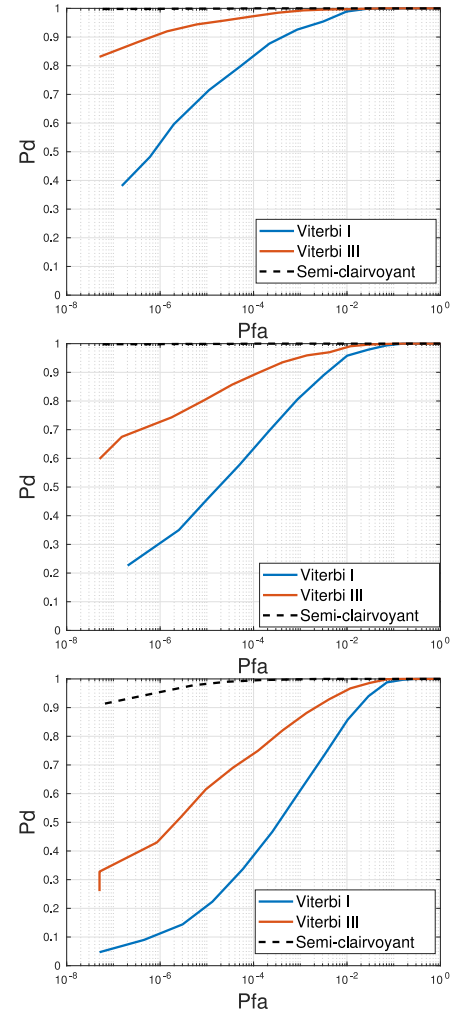


Fig. 3. ROC curve for the HMM with phase-frequency tracking (red curve), the HMM with frequency tracking only (blue curve), and a fully coherent, semi-clairvoyant search (black dashed curve). Detection probability P_d (complement of the false dismissal probability) and false alarm probability P_{fa} are plotted on the vertical and horizontal axes respectively. The black, dashed curve overlaps the upper frame of the figure ($P_d = 1$) in the top and middle panels. Signal parameters: $\mathbf{B}(t) = [(1/3)^{1/2}, (2/3)^{1/2}]$, $\mathbf{A} = h_0[2^{-1/2}, 2^{-1/2}]$, $\lambda = 1.75$ (top), 1.5 (middle), 1.25 (bottom). Search parameters: $T_{\text{drift}} = 10 \text{ d}$, $\gamma T_{\text{drift}} = 8.6 \times 10^{-11}$, $\sigma^2 T_{\text{drift}}^3 = 0.23$, and $N_T = 37$. Realizations: 10^3 per parameter set.

values ≈ 0.2 – 0.5 higher (in absolute terms) than the frequency-only HMM. The corresponding ≈ 1.5 dB improvement in SNR is to be compared with the maximum improvement available from a fully coherent, clairvoyant search, viz. a factor of $N_T^{1/4} \approx 2.5$ or 3.9 dB for $N_T = 37$.

Now consider the number of data blocks or equivalently the drift time-scale. If T_{drift} is fixed, as in Fig. 4(b), then P_d increases, as N_T and hence $T_{\text{obs}} = N_T T_{\text{drift}}$ increase; more data produce better sensitivity. The trend lies somewhere between the asymptotic scalings $\text{SNR} \propto N_T^{1/4}$ and $N_T^{1/2}$ for semi-coherent and fully coherent algorithms respectively. If instead the number of data blocks increases, while T_{obs} is held fixed, P_d behaves as in Fig. 4(c). For $T_{\text{drift}} \lesssim 10 \text{ d}$, P_d increases with T_{drift} , because T_{drift} is less than the wandering time-scale; longer coherent blocks produce better sensitivity. For $T_{\text{drift}} \gtrsim 10 \text{ d}$, P_d decreases with T_{drift} , because T_{drift} exceeds the wandering

²In gravitational wave applications, it is usual to claim a detection, when the Viterbi score surpasses a threshold set by P_{fa} [5], [6], [17].

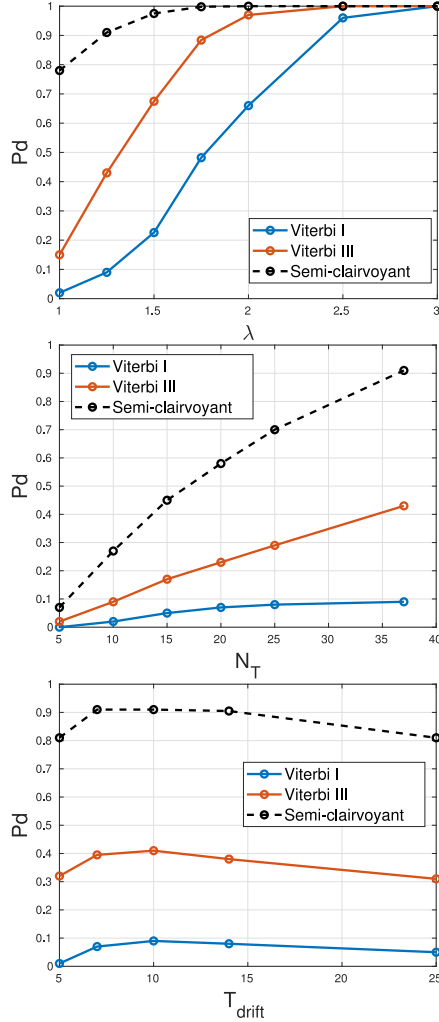


Fig. 4. Detection probability P_d versus (a) squared SNR $\lambda \propto h_0^2$, with $N_T = 37$ and $T_{\text{obs}} = 1$ yr fixed; (b) number of data blocks N_T , with $T_{\text{drift}} = 10$ d fixed; and (c) drift time-scale T_{drift} , with $T_{\text{obs}} = 1$ yr and $\lambda = 1.25$ fixed. The results pertain to a fiducial search covering 20002 frequency bins with $P_{\text{fa}} = 5 \times 10^{-7}$ per bin. Realizations: 10^3 per parameter set.

time-scale, condition (2) is violated, and the transition probabilities in Section II-B become progressively worse approximations. The degradation in P_d is also visible in the semi-clairvoyant control results, as expected. It occurs faster for the phase-frequency HMM — albeit starting from a higher base — than the frequency-only HMM, because φ (the integral of f) is more sensitive to prolonged wandering than f .

V. APPLICATION: GRAVITATIONAL WAVE SEARCHES

As an application, we consider briefly the search for gravitational radiation from cosmic sources. The example is chosen because of (i) its scientific importance, (ii) the astonishing weakness of the signals, which means that gravitational wave detectors operate near the detection limit, where the algorithm in this paper can contribute decisively, and (iii) the fact that HMM frequency tracking without phase tracking has been used successfully in gravitational wave searches before [5], [6], [17].

Gravitational waves are a prediction of Einstein's theory of general relativity [18]. Intuitively they can be imagined as

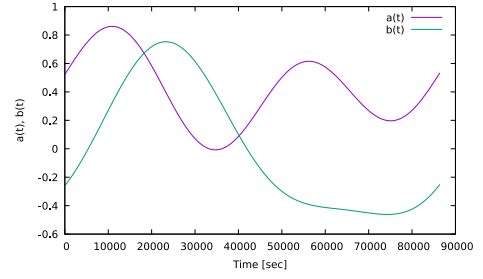


Fig. 5. Dimensionless beam pattern functions $a(t)$ (blue curve) and $b(t)$ (green curve) for the LIGO interferometer located at Livingston, USA, in the direction of the cosmic source Scorpius X-1. The temporal axis is calibrated in seconds; the temporal origin is arbitrary. Parameters: source right ascension $\alpha = 4.27570$ rad, source declination $\delta = -0.27297$ rad, detector latitude 0.5333 rad, detector orientation angle 4.2411 rad (measured counterclockwise from the East to the bisector of the perpendicular interferometer arms).

vibrations of spacetime. More precisely they are traveling-wave perturbations of the metric tensor $g_{\mu\nu}$ describing the coordinate-invariant interval $ds^2 = g_{\mu\nu} dx^\mu dx^\nu$ between two spacetime events displaced infinitesimally by the four-vector dx^μ . In practice, gravitational waves are detected by kilometer-scale, long-baseline, laser interferometers, which are sensitive to arm length changes $\lesssim 10^{-19}$ m. A wave incident normally on the plane of an interferometer stretches and squeezes the arms in a quadrupole pattern [18]. There are two polarizations, named ‘plus’ and ‘cross’, whose principal axes are inclined by $\pi/4$ rad (cf. $\pi/2$ rad for electromagnetic radiation). The leading-order multipole is the quadrupole; dipole radiation is forbidden by momentum conservation.

To date, five direct detections have been reported by the Laser Interferometer Gravitational Wave Observatory (LIGO) of gravitational waves from merging binary black holes and neutron stars [19]–[24]. The mergers emit short-lived chirps lasting ~ 1 s (black holes) and $\sim 10^2$ s (neutron stars) in the decahertz to kilohertz band. In addition, there are sound theoretical reasons to expect persistent, quasimonochromatic gravitational radiation from long-lived cosmic sources like rapidly rotating neutron stars [10]. Such continuous-wave signals are high-priority targets for LIGO. The signal at the detector is a linear combination of two polarizations, whose amplitudes are projected onto the antenna beam-pattern functions $a(t)$ and $b(t)$, which are periodic and incorporate the diurnal rotation of the Earth. The functions $a(t)$ and $b(t)$ are defined in terms of the latitude and orientation of the detector and the sky position (right ascension and declination) of the source via equations (12) and (13) in [15]. They are plotted versus time over a single period (24 hours) in Fig. 5. [The temporal averages in (17)–(19) are constant for $T_{\text{obs}} \geq T_{\text{drift}} \geq 24$ hr to an excellent approximation.] The frequency of the signal is also Doppler shifted by the Earth's rotation and its orbital motion relative to the Solar System barycenter. The signal phase including Doppler corrections is defined in terms of the location of the detector and the sky position and intrinsic frequency evolution of the source via equation (18) in [15].

A full sensitivity study and ROC curve for the gravitational wave application lie outside the scope of this paper and will be reported elsewhere. Among other things, the emission probability (24) needs to be generalized, because the two components

of $\mathbf{R}$ do not share the same phase φ , unlike in (16); they are shifted by an amount that depends on the orientation of the polarization basis of the wave relative to the arms of the detector. As a rough guide, however, it is interesting to scale the Monte Carlo results in Section IV to the gravitational wave problem. Taking $P_{\text{fa}} = 10^{-2}$ and $S_h(f)^{1/2} = 4 \times 10^{-24} \text{ Hz}^{-1/2}$ at $f = 10^2 \text{ Hz}$ (the Advanced LIGO design sensitivity) by way of illustration, we show that the detection probability drops below $P_d = 0.9$ for $h_0 \leq 1 \times 10^{-26}$ with phase-frequency tracking ($\lambda = 1.75$). Here h_0 is the amplitude of the fluctuating part of $g_{\mu\nu}$. The sensitivity is to be compared with theoretical estimates of h_0 for realistic cosmic sources. For example, the accreting neutron star Scorpius X-1, the brightest X-ray source in the sky and a high-priority LIGO target, is estimated to emit gravitational radiation with amplitude $h_0 = 5 \times 10^{-27} (f/0.6 \text{ kHz})^{-1/2} (F_X/10^{-8} \text{ erg cm}^{-2} \text{ s}^{-1})^{1/2}$, based on an angular momentum conservation argument, where F_X is the measured X-ray flux [10].

VI. CONCLUSIONS

In this paper, we extend the standard HMM algorithm for frequency line tracking to track phase as well as frequency. The transition probabilities are constructed to describe a phase-wrapped, Ornstein-Uhlenbeck process during each HMM step, with $f(t)$ and $\varphi(t)$ continuous for all t . The emission probability is a generalized version of a standard, frequency-domain, maximum-likelihood estimator, except that we marginalize rather than maximize the likelihood over the positive-definite amplitude to resolve the phase-amplitude ambiguity uniquely.

Monte Carlo simulations driven by Gaussian noise yield that P_d from phase-tracking exceeds P_d for frequency-only tracking by ≈ 0.10 – 0.30 for $10^{-4} \leq P_{\text{fa}} \leq 10^{-2}$. Expressed another way, for fixed P_d and P_{fa} , phase-frequency tracking detects absolute λ values ≈ 0.2 – 0.5 higher than frequency-only tracking.

The HMM is developed for dual polarizations with slowly varying antenna beam-pattern functions with an eye to applications in gravitational wave astronomy. Modern gravitational wave detectors like Advanced LIGO are astonishingly sensitive, with $S_h(f)^{1/2}$ approaching $4 \times 10^{-24} \text{ Hz}^{-1/2}$, yet still they operate near the detection limit, because the signals are correspondingly weak. The $\approx 1.5 \text{ dB}$ SNR improvement achieved by the algorithm in this paper, although relatively modest, can therefore contribute decisively to gravitational wave searches. The SNR improves $\propto (T_{\text{drift}} T_{\text{obs}})^{1/4}$ for persistent, quasi-monochromatic, gravitational wave signals, as more data are collected, as long as the fluctuating frequency and phase can be tracked semi-coherently; a typical Advanced LIGO observation runs for $T_{\text{obs}} \sim 1 \text{ yr}$, i.e. $\gtrsim 10^9$ cycles for a hectohertz source. For this reason, HMM-based algorithms like the one studied here, which detect and track a wandering tone whose parameters are unknown a priori from electromagnetic observations, are genuinely valuable.

APPENDIX

PHASE-WRAPPED ORNSTEIN-UHLENBECK PROCESS

In this paper, we model the frequency-phase evolution during an HMM step $t_n \leq t \leq t_{n+1}$ as an Ornstein-Uhlenbeck process

defined by the stochastic differential equations (3) and (4). The process is identical to traditional Brownian motion, with “velocity” and “displacement” variables f and φ , except that it is periodic in φ [25]. In this Appendix, we solve the forward and backward Fokker-Planck equations corresponding to (3) and (4) for the PDF and first- and second-order moments. From these quantities, it is straightforward to calculate the HMM transition probabilities.

Consider a general random process described by a state variable $\mathbf{q}(t)$ governed by the stochastic differential equation

$$\frac{d\mathbf{q}_i}{dt} = C_i[\mathbf{q}(t)] + D_{ij}[\mathbf{q}(t)]\xi_j(t), \quad (26)$$

with $\langle \xi_i(t) \rangle = 0$ and $\langle \xi_i(t)\xi_j(t') \rangle = \delta_{ij}\delta(t-t')$, where C_i and D_{ij} are state-dependent drift and diffusion coefficients respectively. The forward PDF $p^F(t, \mathbf{q}) = p[t, \mathbf{q}|t_n, \mathbf{q}(t_n)]$ corresponding to (26) satisfies the forward Fokker-Planck equation

$$\frac{\partial p^F}{\partial t} = -\frac{\partial(C_i p^F)}{\partial q_i} + \frac{1}{2} \frac{\partial^2(D_{ik}D_{jk} p^F)}{\partial q_i \partial q_j} \quad (27)$$

for $t \geq t_n$. The backward PDF $p^B(t, \mathbf{q}) = p[t, \mathbf{q}|t_{n+1}, \mathbf{q}(t_{n+1})]$ corresponding to (26) satisfies the backward Fokker-Planck equation

$$-\frac{\partial p^B}{\partial t} = C_i \frac{\partial p^B}{\partial q_i} + \frac{1}{2} D_{ik} D_{jk} \frac{\partial^2 p^B}{\partial q_i \partial q_j} \quad (28)$$

for $t \leq t_{n+1}$. For the phase-wrapped, Ornstein-Uhlenbeck process described by (3) and (4), we have $C_1 = -\gamma q_1$, $C_2 = q_1$, $D_{11} = \sigma$, $D_{12} = D_{21} = D_{22} = 0$, and hence

$$\frac{\partial p^F}{\partial t} = \gamma p^F + \gamma f \frac{\partial p^F}{\partial f} - f \frac{\partial p^F}{\partial \varphi} + \frac{\sigma^2}{2} \frac{\partial^2 p^F}{\partial f^2} \quad (29)$$

and

$$\frac{\partial p^B}{\partial t} = \gamma f \frac{\partial p^B}{\partial f} - f \frac{\partial p^B}{\partial \varphi} - \frac{\sigma^2}{2} \frac{\partial^2 p^B}{\partial f^2}. \quad (30)$$

Upon multiplying (29) by the integrating factor $\exp(-\gamma t)$, we find that (29) reduces to (30) with σ^2 replaced by $-\sigma^2$ and the initial conditions of the forward PDF replaced by the final conditions of the backward PDF, i.e. we have

$$p^B(t, f, \varphi) \propto \exp(-\gamma t) p^F(t, f, \varphi)|_{\sigma^2 \mapsto -\sigma^2}, \quad (31)$$

with the proportionality constant determined by normalization.

We solve (29) by expanding $p^F(t, f, \varphi)$ as a Fourier series in φ (periodic boundary condition) and a Fourier transform in f (absorbing boundary condition); we have $p^F \rightarrow 0$ as $f \rightarrow \pm\infty$ in principle, although in practice p^F peaks narrowly within one bin of $f(t_n)$ by construction in the HMM application treated here. Writing

$$p^F(t, f, \varphi) = (2\pi)^{-2} \sum_{n=-\infty}^{\infty} \exp(in\varphi) \int_{-\infty}^{\infty} dk \times \exp(ikf) \tilde{p}^F(t, n, \kappa), \quad (32)$$

with

$$\begin{aligned} \tilde{p}^F(t, n, \kappa) &= \int_0^{2\pi} d\varphi \int_{-\infty}^{\infty} df \\ &\times \exp(-in\varphi - i\kappa f) p^F(t, f, \varphi), \end{aligned} \quad (33)$$

we find that the characteristic function $\tilde{p}^F(t, n, \kappa)$ satisfies the first-order partial differential equation

$$\frac{\partial \tilde{p}^F}{\partial t} = (-\gamma\kappa + n) \frac{\partial \tilde{p}^F}{\partial \kappa} - \frac{\sigma^2 \kappa^2 \tilde{p}^F}{2}, \quad (34)$$

subject to the initial condition

$$\tilde{p}^F(t = 0, n, \kappa) = \exp[-in\varphi(t_n)] \exp[-i\kappa f(t_n)]. \quad (35)$$

Equation (34) can be solved, by the method of characteristics or otherwise, to give

$$\begin{aligned} \tilde{p}^F(t, n, \kappa) &= \exp \left[-in\varphi(t_n) - i\rho f(t_n) \right. \\ &+ \frac{\sigma^2}{4\gamma} \left(\rho - \frac{n}{\gamma} \right) \left(\rho + \frac{3n}{\gamma} \right) \exp \left\{ -\frac{\sigma^2}{2} \left[\frac{n^2 \tau}{\gamma^2} \right. \right. \\ &+ \frac{2n}{\gamma^2} \left(\rho - \frac{n}{\gamma} \right) \exp(\gamma\tau) \\ &\left. \left. + \frac{1}{2\gamma} \left(\rho - \frac{n}{\gamma} \right)^2 \exp(2\gamma\tau) \right] \right\} \left. \right] \end{aligned} \quad (36)$$

with

$$\rho = \frac{n}{\gamma} + \left(\kappa - \frac{n}{\gamma} \right) \exp(-\gamma\tau) \quad (37)$$

and $\tau = t - t_n$. Equation (37) can be rewritten explicitly as a Gaussian function by completing the square in the curly braces in (36). It reduces to the standard Ornstein-Uhlenbeck result without phase wrapping for $n = 0$ [25]. Substituting (36) and (37) into (32) and thence into (31) gives (7)–(12) in Section II-B.

The moments are calculated directly from the characteristic function via

$$\langle f^l \varphi^m \rangle = i^{l+m} \frac{\partial^{l+m} \tilde{p}^F}{\partial \kappa^l \partial n^m} \Big|_{(\kappa, n) = (0, 0)}. \quad (38)$$

The results up to second order are

$$\langle f \rangle = f(t_n) \exp(-\gamma\tau), \quad (39)$$

$$\langle \varphi \rangle = \varphi(t_n) + \frac{f(t_n)}{\gamma} [1 - \exp(-\gamma\tau)], \quad (40)$$

$$\langle (f - \langle f \rangle)^2 \rangle = \frac{\sigma^2}{2\gamma} [1 - \exp(-2\gamma\tau)], \quad (41)$$

$$\langle (f - \langle f \rangle)(\varphi - \langle \varphi \rangle) \rangle = \frac{\sigma^2}{2\gamma^2} [1 - \exp(-\gamma\tau)]^2, \quad (42)$$

$$\langle (\varphi - \langle \varphi \rangle)^2 \rangle = \frac{\sigma^2}{2\gamma^3} \{1 + 2\gamma\tau - [2 - \exp(-\gamma\tau)]^2\}. \quad (43)$$

The moments of the forward and backward PDFs are the same, except that the initial and final conditions are switched, as discussed above.

REFERENCES

- [1] R. L. Streit and R. F. Barrett, "Frequency line tracking using hidden Markov models," *IEEE Trans. Acoust., Speech, Signal Process.*, vol. 38, no. 4, pp. 586–598, Apr. 1990.
- [2] B. G. Quinn and E. J. Hannan, *The Estimation and Tracking of Frequency*. Cambridge, U.K.: Cambridge Univ. Press, 2001.
- [3] S. Paris and C. Jauffret, "Frequency line tracking using HMM-based schemes," *IEEE Trans. Aerosp. Electron. Syst.*, vol. 39, no. 2, pp. 439–449, Apr. 2003.
- [4] L. B. White and R. J. Elliott, "A mixed MAP/MLSE receiver for convolutional coded signals transmitted over a fading channel," *IEEE Trans. Signal Process.*, vol. 50, no. 5, pp. 1205–1214, May 2002.
- [5] S. Suvorova, L. Sun, A. Melatos, W. Moran, and R. Evans, "Hidden Markov model tracking of continuous gravitational waves from a neutron star with wandering spin," *Phys. Rev. D*, vol. 93, no. 12, 2016, Art. no. 123009.
- [6] S. Suvorova, P. Clearwater, A. Melatos, L. Sun, W. Moran, and R. Evans, "Hidden Markov model tracking of continuous gravitational waves from a binary neutron star with wandering spin. ii. Binary orbital phase tracking," *Phys. Rev. D*, vol. 96, no. 10, 2017, Art. no. 102006.
- [7] X. Xie and R. J. Evans, "Multiple target tracking and multiple frequency line tracking using hidden Markov models," *IEEE Trans. Signal Process.*, vol. 39, no. 12, pp. 2659–2676, Dec. 1991.
- [8] R. F. Barrett and D. A. Holdsworth, "Frequency tracking using hidden Markov models with amplitude and phase information," *IEEE Trans. Signal Process.*, vol. 41, no. 10, pp. 2965–2976, Oct. 1993.
- [9] X. Xie and R. J. Evans, "Frequency-wavenumber tracking using hidden Markov models," *IEEE Trans. Signal Process.*, vol. 41, no. 3, pp. 1391–1394, Mar. 1993.
- [10] K. Riles, "Gravitational waves: Sources, detectors and searches," *Prog. Part. Nucl. Phys.*, vol. 68, pp. 1–54, Jan. 2013.
- [11] L. R. Rabiner, "A tutorial on hidden Markov models and selected applications in speech recognition," in *Readings in Speech Recognition*. New York, NY, USA: Elsevier, 1990, pp. 267–296.
- [12] G. E. Uhlenbeck and L. S. Ornstein, "On the theory of the Brownian motion," *Phys. Rev.*, vol. 36, no. 5, pp. 823–841, 1930.
- [13] B. Øksendal, "Stochastic differential equations," in *Stochastic Differential Equations*. Berlin, Germany: Springer, 2003, pp. 65–84.
- [14] D. R. A. McMahon and R. F. Barrett, "Generalization of the method for the estimation of the frequencies of a tone in noise from the phases of discrete Fourier transforms," *IEEE Trans. Signal Process.*, vol. SP-12, no. 4, pp. 317–383, Jun. 1987.
- [15] P. Jaranowski, A. Krolak, and B. F. Schutz, "Data analysis of gravitational-wave signals from spinning neutron stars: The signal and its detection," *Phys. Rev. D*, vol. 58, no. 6, 1998, Art. no. 063001.
- [16] R. Prix and B. Krishnan, "Targeted search for continuous gravitational waves: Bayesian versus maximum-likelihood statistics," *Classical Quantum Gravity*, vol. 26, no. 20, Oct. 2009, Art. no. 204013.
- [17] The LIGO Scientific Collaboration, the Virgo Collaboration, B. P. Abbott, R. Abbott, T. D. Abbott, F. Acernese, K. Ackley, C. Adams, T. Adams, P. Addesso, and *et al.*, "Search for gravitational waves from Scorpius X-1 in the first Advanced LIGO observing run with a hidden Markov model," *Phys. Rev. D*, vol. 95, Jun. 2017, Art. no. 122003.
- [18] C. W. Misner, K. S. Thorne, and J. A. Wheeler, *Gravitation*. San Francisco, CA, USA: Freeman, 1973, p. 891.
- [19] B. P. Abbott *et al.*, "Observation of gravitational waves from a binary black hole merger," *Phys. Rev. Lett.*, vol. 116, no. 6, Feb. 2016, Art. no. 061102.
- [20] B. P. Abbott *et al.*, "GW151226: Observation of gravitational waves from a 22-solar-mass binary black hole coalescence," *Phys. Rev. Lett.*, vol. 116, no. 24, Jun. 2016, Art. no. 241103.
- [21] B. P. Abbott *et al.*, "GW170104: Observation of a 50-solar-mass binary black hole coalescence at redshift 0.2," *Phys. Rev. Lett.*, vol. 118, no. 22, Jun. 2017, Art. no. 221101.
- [22] B. P. Abbott *et al.*, "GW170814: A three-detector observation of gravitational waves from a binary black hole coalescence," *Phys. Rev. Lett.*, vol. 119, no. 14, Oct. 2017, Art. no. 141101.
- [23] B. P. Abbott *et al.*, "GW170817: Observation of gravitational waves from a binary neutron star inspiral," *Phys. Rev. Lett.*, vol. 119, no. 16, Oct. 2017, Art. no. 161101.

- [24] The LIGO Scientific Collaboration, the Virgo Collaboration, B. P. Abbott, R. Abbott, T. D. Abbott, F. Acernese, K. Ackley, C. Adams, T. Adams, P. Addesso, and *et al.*, “GW170608: Observation of a 19-solar-mass Binary Black Hole Coalescence,” *Astrophys. J. Lett.*, vol. 851, 2017, Art. no. L35.
- [25] C. W. Gardiner, *Handbook of Stochastic Methods for Physics, Chemistry and the Natural Sciences*. Berlin, Germany: Springer-Verlag, 1994.



Sofia Suvorova received the Ph.D. degree in mathematics from Flinders University, Adelaide, SA, Australia. She has worked across a range of projects in wavelets, signal processing, optimisation, radar and sensor management. After two years as a Research Scientist in Defence Science and Technology Group, she was with the Department of Electrical and Electronic Engineering, University of Melbourne for more than 10 years. Her current interests include signal processing for gravitational waves and optimisation of dynamical systems.



Andrew Melatos received the B.Sc. and Ph.D. degrees from the University of Sydney, Sydney, NSW, Australia, in 1995. He is currently a Theoretical Physicist with the University of Melbourne, Parkville, VIC, Australia. He held Postdoctoral Fellowships with the California Institute of Technology (1995–1997) and the University of California at Berkeley (1997–2000). His research interests include plasma physics, high-energy astrophysics (with a focus on neutron stars), gravitational physics, nonequilibrium statistical mechanics, condensed matter

physics, and econophysics. He is currently a member of the Laser Interferometer Gravitational Wave Observatory (LIGO) Scientific Collaboration and a Node Leader of the Australian Research Council Centre of Excellence for Gravitational Wave Discovery (OzGrav). With his LIGO colleagues, he is a joint recipient of the Special Breakthrough Prize in Fundamental Physics, awarded in 2016 for the first direct detection of gravitational waves.



Rob J. Evans (F'02) was born in Melbourne, VIC, Australia, in 1947. He received the B.E. degree in electrical engineering from the University of Melbourne, Parkville, VIC, Australia, in 1969 and the Ph.D. degree from the University of Newcastle, Callaghan, NSW, Australia, in 1975. He was a Radar Systems Engineering Officer with the Royal Australian Airforce. After completing the Ph.D. degree, he did postdoctoral studies at the Laboratory for Information and Decision Systems, MIT, USA, and the Control and Management Department,

Cambridge University, Cambridge, U.K. In 1977, he took up an Academic Position with the University of Newcastle, where he was the Head of the Department of Electrical and Computer Engineering from 1986 to 1991 and the Chief Investigator and Co-Director of the ARC Centre of Excellence on Industrial Control Systems between 1988–1991. In 1992, he joined the University of Melbourne, where he has served in many roles including Head of the Department of Electrical and Electronic Engineering for the periods 1993–1996 and 2013–2017, a Research Leader for the Cooperative Research Centre for Sensor Signal and Information Processing 1992–2000, the Director of the DSTO Centre for Networked Decision and Sensor Systems 2001–2004, the Director of the Victoria Research Laboratory of National ICT Australia 2004–2012, an Executive Dean of Engineering during 2007, and the Director of the Defence Sciences Institute 2014–2017. He is currently a Melbourne University Laureate Professor and a Chief Investigator in the ARC Centre of Excellence for Gravity Wave Detection. He was on the IFAC Council from 2002 to 2008. His research and industry engagement has ranged across many areas including theory and applications in control systems, industrial electronics, radar systems, signal processing, and telecommunications. He is currently a Fellow of the Australian Academy of Science, a Fellow of the Australian Academy of Technological Sciences and Engineering, and a Fellow of the Institution of Engineers Australia.

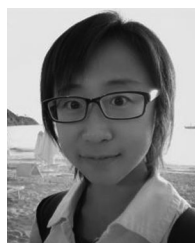


William Moran (M'95) received the B.Sc. (with first class Hons.) degree in mathematics from the University of Birmingham, Birmingham, U.K., in 1965 and the Ph.D. degree in pure mathematics from the University of Sheffield, Sheffield, U.K. in 1968. Since 2017, he has been a Professor of defence technology with the University of Melbourne, Parkville, VIC, Australia. From 2014 to 2017, he was the Director of the Signal Processing and Sensor Control Group, the School of Engineering, RMIT University, from 2001 to 2014, a Professor with the Department of Electrical

Engineering, University of Melbourne, the Director with the Defence Science Institute, University of Melbourne (2011–2014), a Professor of mathematics (1976–1991), the Head of the Department of Pure Mathematics (1977–1979, 1984–1986), the Dean of mathematical and computer sciences (1981, 1982, 1989), the University of Adelaide, and the Head of the mathematics discipline, the Flinders University of South Australia (1991–1995). He was the Head of the Medical Signal Processing Program (1995–1999) in the Cooperative Research Centre for Sensor Signal and information Processing. His main areas of research interests include signal processing both theoretically and in applications to radar, waveform design and radar theory, sensor networks, and sensor management. He also works in various areas of mathematics including harmonic analysis, representation theory, and number theory. He was a member of the Australian Research Council College of Experts from 2007 to 2009. He was elected to the Fellowship of the Australian Academy of Science in 1984. He was a Principal Investigator on numerous research grants and contracts, in areas spanning pure mathematics to radar development, from both Australian and U.S. Research Funding Agencies, including DARPA, AFOSR, AFRL, Australian Research Council (ARC), Australian Department of Education, Science and Training, and Defence Science and Technology, Australia.



Patrick Clearwater is currently working toward the Ph.D. degree at the University of Melbourne, Parkville, VIC, Australia. He is currently with the School of Physics and with the ARC Centre of Excellence for Gravitational Wave Discovery (OzGrav). His current research interests include data analysis to search for continuous-wave signals from spinning neutron stars in particular. In 2018, she started working as a Postdoctoral Scholar with the California Institute of Technology, continuing her research interests include gravitational-wave physics. Her primary research interests include gravitational-wave signal processing and learning fundamental



Ling Sun received the Ph.D. degree in physics from the University of Melbourne, Parkville, VIC, Australia. She has been working on gravitational-wave data analysis within the LIGO Scientific Collaboration, searching for continuous-wave signals from spinning neutron stars in particular. In 2018, she started working as a Postdoctoral Scholar with the California Institute of Technology, continuing her research interests include gravitational-wave physics. Her primary research interests include gravitational-wave signal processing and learning fundamental

physics from gravitational-wave observations.